

PROJECT REPORT: IRON EYES

Integrated Velocity-Based Training (VBT), Load Autoregulation, and Movement Feedback System

Smart India Hackathon 2026

- **Problem Statement ID:** SIH26213
- **Problem Statement Title:** Student Innovation — Ideas that can boost fitness activities and assist in keeping fit
- **Theme:** Fitness & Sports
- **PS Category:** Hardware
- **Team Name:** Iron Eyes
- **Repository & Documentation:** [GitHub - Iron Eyes](#)

1. Executive Summary & Problem Analysis

The Injury Problem in Strength Sports

In resistance training, strength sports, and high-performance athletics, non-contact overuse injuries and high re-injury rates stem primarily from poor load management, unmonitored central nervous system (CNS) fatigue, and poor movement execution.

INJURY INCIDENCE BY NUMBERS	
Bodybuilding	: 0.9 injuries / 1,000 training hours
Olympic Weightlifting	: 3.0 injuries / 1,000 training hours
Powerlifting	: 3.3 injuries / 1,000 training hours
Prior Injury Risk	: 3.75x higher re-injury risk if unmanaged

- **Powerlifting:** 63.9% of powerlifters report a career acute injury (Stronger by Science survey, $n=1,543$).
- **Weightlifting:** 27% of weightlifters report being injured within the past 6 months (393-lifter survey, 2023).
- **Comparative Rates:** Incidence reaches 3.0 (2.4–3.6) injuries per 1,000 hours in Olympic Weightlifting and 3.3 per 1,000 hours in Powerlifting, compared to 0.9 (0.8–1.0) in Bodybuilding.
- **Re-injury Cascade:** Cross-sectional reviews indicate a **3.75x higher re-injury risk** following an unmanaged prior injury.

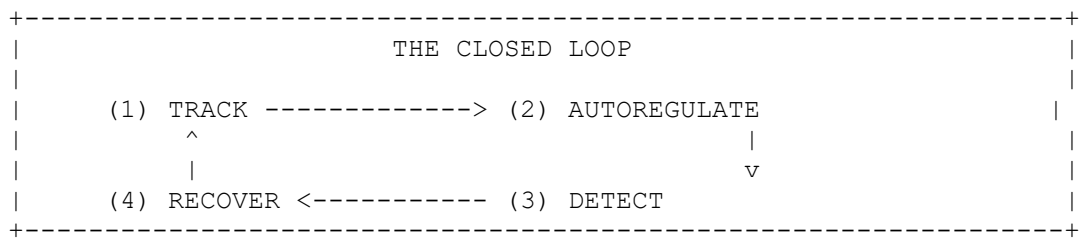
Core Innovation

Iron Eyes addresses these challenges by transforming every working set into objective kinematic data. Combining a low-cost, magnetic wearable hardware unit with a cross-platform processing

application, the system measures movement velocity, controls fatigue, autoregulates training loads, and connects persistent movement degradation to structured recovery protocols.

2. The 4-Stage Closed-Loop Architecture

Iron Eyes operates as a fully closed-loop ecosystem where every completed repetition informs subsequent load decisions.

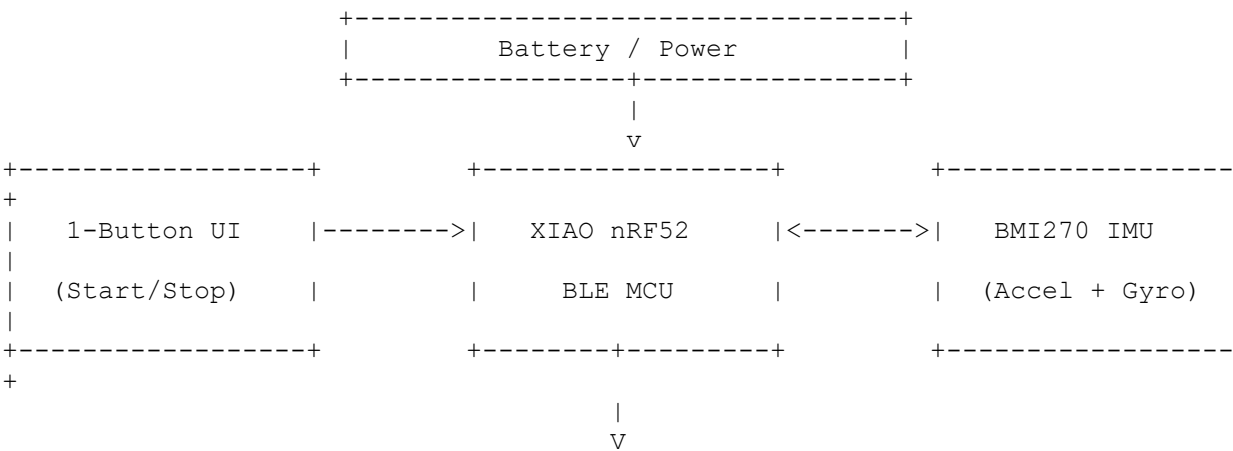


- 1. **TRACK:** Continuous barbell velocity and path monitoring via IMU data fusion streaming real-time metrics over Bluetooth Low Energy (BLE).
- 2. **AUTOREGULATE:** Real-time dynamic fatigue assessment that adapts load and volume prescriptions based on daily performance metrics (e1RM, load-velocity profiles) to prevent overtraining.
- 3. **DETECT:** Bar path and technique analysis compared against person-specific baselines to detect dangerous movement compensations, delivering real-time haptic and app alerts.
- 4. **RECOVER:** Longitudinal performance analysis and return-to-play decisions powered by objective recovery tracking and personalized protocol prescriptions.

3. System Hardware Architecture & Firmware Pipeline

Hardware Components

The wearable unit mounts directly to the barbell using a magnetic attachment, requiring zero manual app intervention mid-set.





- **Microcontroller Unit (MCU):** Seeed Studio XIAO nRF52 (Nordic nRF52 series) handling on-device sampling, sensor control, and wireless streaming.
- **Inertial Measurement Unit (IMU):** Bosch BMI270 6-axis accelerometer and gyroscope capturing high-frequency kinematic movement.
- **User Interface & Form Factor:** Single physical push button for set control (start/stop) with magnetic casing for quick barbell clipping.

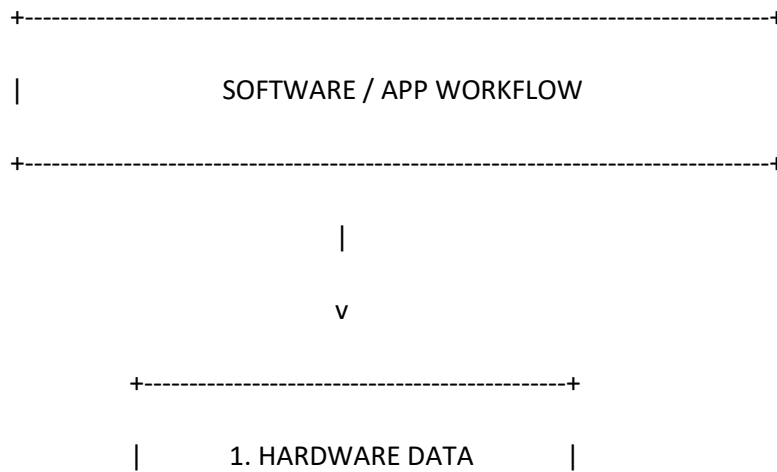
Firmware & Data Processing Stack

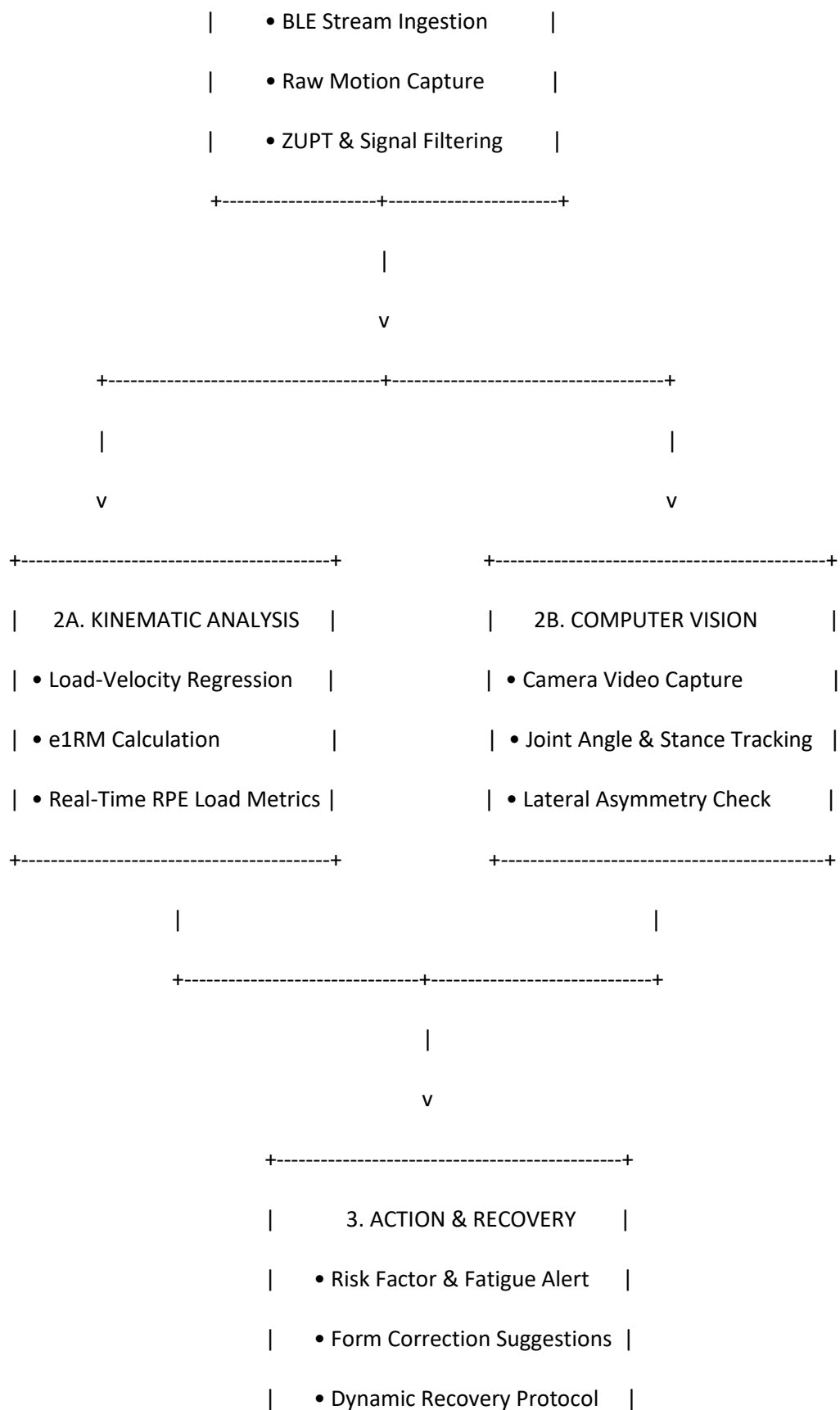
- **OS & Kernel Environment:** Zephyr RTOS, Nordic nRF Connect SDK, and Linux kernel tools for task scheduling and hardware abstraction.
- **Kinematics Engine:** Signal filtering (low-pass) combined with Zero-Velocity Updates (ZUPT) to prevent digital integration drift during inter-rep stationary phases.
- **BLE Communications:** Low-latency packet transmission transferring raw and pre-processed motion frames directly to the app.

4. Software Architecture & Intelligence Engines

Mobile & Computer Vision Stack

- **Application Framework:** Cross-platform mobile development using Flutter / Node.js.
- **Computer Vision Pipeline:** Python-backed vision modules employing OpenCV, MediaPipe, NumPy, and Pandas for pose estimation, joint angle extraction, and bar-path asymmetry tracking.
- **AI & Diagnostic Layer:** Integration options for deterministic logic engines supported by LLM reasoning frameworks (Claude AI / OpenAI API).







Motion Tracking Accuracy & Drift Control (ZUPT Evaluation)

Integration of acceleration data natively suffers from baseline velocity drift. Iron Eyes implements a Zero-Velocity Update (ZUPT) algorithm triggered when stationary states are detected at movement lockouts.

Metric / Parameter	Standard IMU (Without ZUPT)	Iron Eyes IMU (With ZUPT)
Velocity Drift Rate	$\sim 0.4 \text{ m/s per rep}$	$< 0.03 \text{ m/s per rep}$
e1RM Estimation Error	32.5%	5.5%
Signal Stability	Accumulative Integration Shift	Corrected Drift at Lockout Intercepts

5. Feasibility, Risk Management, & Development Roadmap

Feasibility Pillars

- Sensor Fusion Stability:** Standardized low-latency filtering combined with ZUPT yields stable, repeatable linear velocity measurements.
- Validated Sports Science:** Load-velocity profiles and estimated 1RM calculations leverage peer-reviewed biomechanical equations rather than unpredictable black-box models.
- Low System Overhead:** Deterministic rule-based engines reduce cloud dependencies, GPU infrastructure costs, and continuous AI model retraining.

Risk Matrix & Mitigations

Risk Category	Identified Vulnerability	Actionable Mitigation Strategy
IMU Drift	Accumulative velocity drift over time.	Apply ZUPT drift correction paired with multi-sensor fusion filters.

Risk Category	Identified Vulnerability	Actionable Mitigation Strategy
Overclaiming AI	Unexplainable outputs from black-box models.	Utilize deterministic linear regression and transparent, rule-based algorithms.
Scope Creep	Development delays during tight timelines.	Strict feature freezing and isolation of milestone tasks.
Rehab Liability	Risk of medical misdiagnosis claims.	Embedded "Coach-Assist, Not Diagnosis" guidelines and disclaimers.
Bandwidth Limits	Module cross-dependency bottlenecks.	Single technical owner assignment for each module.

Development Phases

PHASE 0: Hardware & Velocity Telemetry
 - MPU6050/BMI270 sensor setup, C++/Arduino firmware, BLE 5.0 streaming, velocity algorithms.

PHASE 1: Load Engine & Core Dashboard
 - Python/FastAPI load-velocity regression profiles, REST APIs, React athlete/coach UI.

PHASE 2: Application & Recovery Engine
 - Clinical rule engine integration, SQLite/PostgreSQL, automated rehab prescriptions.

PHASE 3: Computer Vision Asymmetry
 - OpenCV & MediaPipe video parsing for joint angles, lateral shift, and stance analysis.

6. Competitive Advantage & Impact

Product Comparison Matrix

Product	Mechanism / Tech	Key Limitations	Iron Eyes Advantage
GymAware / Output Sports	Linear encoder / tether hardware.	High cost (\$\geq \\$2,000+\$); lacks integrated rehab feedback.	3\$\times\$ to 5\$\times\$ cheaper indigenous hardware.
Vmaxpro	Wireless Bluetooth IMU.	Load tracking only; lacks automated rehab engine.	Combines velocity tracking with auto-recovery logic.
Stance Fitness / Lift.app.ai	Smartphone camera form tracking.	Highly sensitive to camera placement; zero bar velocity data.	Combines camera pose tracking with direct hardware IMU data.
Iron Eyes	IMU + Vision + Autoregulation + Rehab	<i>None (Unified Ecosystem)</i>	First complete, low-cost, closed-loop sports science system.

Target Stakeholders & Multi-Dimensional Impact

- **Strengthening & Conditioning (S&C) Coaches:** Automated daily load adjustments and team fatigue tracking dashboards.
- **Athletes & Lifters:** Real-time velocity feedback, target RPE accuracy, PR safety checks, and burnout prevention.
- **Physiotherapists:** Objective velocity metrics and joint angle diagnostics for return-to-play protocols.
- **Socio-Economic Value:** Delivers elite sports science tools at a fraction of the cost of imported devices, supporting grassroots academies, sports institutes, and national fitness initiatives.

7. Research References

1. **Ruf, L., Chery, C., & Taylor, K. (2018).** *Validity and Reliability of the Load-Velocity Relationship to Predict the One-Repetition Maximum in Deadlift.* Journal of Strength and Conditioning Research, 32(3), 681-689.
2. **Strength and Conditioning Journal (2025).** *A Coach's Guide to Velocity-Based Training: Definitions and Diagnostics.* Strength and Conditioning Journal, 47(2), 230-244.

3. **Ma, M., Song, Q., Li, Y., & Zhou, Z. (2018).** *An Adaptive Zero Velocity Detection Algorithm Based on Multi-Sensor Fusion for a Pedestrian Navigation System.* *Sensors*, 18(10), 3261.
4. **Lugaresi, C., et al. (2019).** *MediaPipe: A Framework for Building Perception Pipelines.* arXiv preprint arXiv:1906.08172.
5. Commercial benchmark pricing and feature data referenced from public specifications (GymAware, Output Sports, Vmaxpro, 2026).